

FLOW.OS

WhatsApp AI Platform — Nigerian SMEs
Build Plan, Cost Breakdown & Monthly Running Costs

I. The Short Answer

Two numbers. That's all you need to know upfront.

Item	Cost
Build Fee (one-time — everything gets built)	₦3,500,000
Monthly Running Cost per Business (Chatwoot + AWS + Meta fees)	₦60,000 – ₦76,000 / month

These are two completely separate things. The ₦3.5M is my one-time fee to build the whole platform. The monthly figure is what it costs to keep it running — cloud hosting, AI, and the WhatsApp inbox tool.

Previous estimate used 360dialog at ₦79,500/business/month. New architecture uses Chatwoot + Meta Cloud API direct — bringing that cost down to ~₦28,500/business/month. The platform is now significantly more affordable.

2. How It Works — The Architecture

FlowOS runs on two tools that each business owner uses daily.

Tool	What It's For	Who Uses It
Chatwoot Inbox	Replying to customers, confirming orders, taking over from the AI	Business owner + staff — this is the customer-facing inbox
FlowOS Dashboard	Revenue analytics, product catalogue, AI settings, staff management	Business owner — for running and configuring the business

The AI does all the heavy lifting — answering FAQs, collecting orders, handling bookings. Chatwoot only comes into play when a human needs to step in or confirm an order. Everything is connected in the background through your AWS backend.

How an Order Works (Step by Step)

Step	What Happens
1	Customer messages the business WhatsApp number
2	Meta Cloud API → Chatwoot → your AWS Lambda backend
3	AI classifies intent, searches products, generates reply → sends back via Chatwoot
4	Customer says 'I want make an order'
5	AI collects order details, calculates total, saves order to database
6	Owner gets a WhatsApp alert on their personal number: '🔔 New order pending '
7	Owner taps the link → opens Chatwoot inbox → sees Confirm / Cancel buttons
8	Owner taps Confirm → customer receives: 'Order confirmed! Delivery by 7pm'
9	Order status updates live in the FlowOS dashboard analytics

3. What We're Building

Delivered in 5 milestones. Each one ends with something working that you can see and test.

Milestone	What Gets Built	What You Can Do After
M1 Foundation	Backend setup, PostgreSQL + PGVector database, authentication, Chatwoot Cloud deployment, WhatsApp Cloud API connection (embedded signup), webhook routing between Meta → Chatwoot → Lambda, deployment pipeline on AWS	Messages flow from WhatsApp through Chatwoot into your backend. The system is live and receiving.
M2 AI Engine	Bedrock AI integration (Nova Lite for classification, Claude Sonnet for responses, Titan for embeddings), PGVector product search, conversation memory in Redis, human takeover logic (AI pause/resume via Chatwoot API), escalation threshold settings	AI handles real conversations. Staff can take over any chat from Chatwoot with one click. AI resumes when they're done.
M3 Orders & Leads	Order collection flow (AI extracts items, quantity, address, total), booking/reservation flow, lead capture and scoring, order confirmation buttons in Chatwoot (Confirm / Cancel / Ask details), owner WhatsApp notification alerts, customer order history	Customer places order → AI captures it → owner gets alerted → confirms in Chatwoot → customer gets confirmation. Full order loop working.
M4 Dashboard	FlowOS business dashboard (React): revenue analytics, order history and status, product catalogue editor, FAQ/knowledge base editor, AI behaviour settings, staff management + Chatwoot access control, notification preferences	Business owner has a full command center — analytics, products, AI config — separate from the Chatwoot inbox.

M5 Launch	Security hardening, stress testing, performance tuning, error handling and logging (CloudWatch), production deployment, onboarding documentation for new businesses, multi-business routing proxy (phone_number_id), final QA across all 5 pilot businesses	A stable, secure product ready for real customers. All 5 businesses live.
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4. Monthly Running Costs — Full Breakdown

Based on 5 businesses, each handling ~5 customer conversations per day (150/month). Here is exactly where every naira goes.

Part A — AWS Cloud Infrastructure

Shared across all 5 businesses. Divided evenly per business below.

Service	What It Does	Monthly Requests	USD / month	NGN / month
Lambda	Processes every message, calls the AI, saves to database	5,400 invocations	\$0.00	₦0
API Gateway	Receives WhatsApp webhooks from Chatwoot + handles dashboard API calls	5,400 calls	~\$0.02	~₦30
RDS PostgreSQL + PGVector	Stores all data — messages, orders, leads, products, embeddings. PGVector powers smart product search at no extra charge	~4,950 ops	~\$11.11	~₦16,665
ElastiCache Redis	Remembers conversation context so the AI knows what was said earlier in a chat	Flat rate	~\$11.68	~₦17,520
S3	Stores uploaded files, logs, and daily backups	~300 requests	~\$0.12	~₦180
Amazon Bedrock	Claude Sonnet handles responses, Nova Lite classifies intent, Titan Embeddings power product search	~600 AI calls + 70 embeddings	~\$4.59	~₦6,885
CloudWatch + Secrets Manager + Route 53	Monitoring, secure key storage, domain DNS	Background	~\$6.00	~₦9,000
AWS Total (5 businesses)			~\$33.52	~₦50,280
÷ 5 businesses = cost per business			~\$6.70	~₦10,050

Part B — External Services (Per Business)

Service	What It Does	Cost Model	NGN / month
Chatwoot Cloud (Startups)	The human inbox — business owner replies to escalated chats and confirms orders here. Comes with conversation history, labels, and quick-reply buttons.	\$19 per agent seat per business	~₦28,500
Meta Conversation Fees	Fee Meta charges per conversation. Customer-initiated (service) conversations are low cost. Estimated conservatively.	~150 convos × ~\$0.01	~₦1,500–₦15,000
Domain (flowos.app)	Custom domain for the FlowOS dashboard	~\$12/year ÷ 12 ÷ 5 businesses	~₦25
Vercel (Dashboard Hosting)	Hosts the FlowOS React dashboard. Free tier covers MVP testing.	Free Tier	₦0

Grand Total Per Business Per Month

Category	Cost (NGN)
AWS Infrastructure (shared, split per business)	~₦10,050
Chatwoot (1 agent seat)	~₦28,500
Meta Conversation Fees	~₦1,500–₦15,000
Domain + Vercel	~₦25
TOTAL PER BUSINESS / MONTH	~₦60,000 – ₦76,000

5 businesses total = ~~~₦300,000–₦380,000~~/month in running costs. This is operational cost, completely separate from the ₦3.5M build fee. You would charge each business a monthly subscription — the gap between what you charge and what it costs is your margin.

5. Why This Is Cheaper Than the Previous Plan

	Old Plan (360dialog)	New Plan (Chatwoot + Meta Direct)
WhatsApp BSP fee	€49/number/month (~₦79,500)	\$0 — connect directly to Meta
Human inbox	Had to build custom dashboard	Chatwoot — ready-made at \$19/seat
Cost per business / month	~₦79,500 (WhatsApp fee alone)	~₦38,550 (Chatwoot + AWS)

Saving per business / month

~~~₦41,000~~ saved per business

## 6. How Costs Change as You Grow

| Businesses | Chatwoot Total          | AWS Total (shared)    | Total / month            | Cost / business        |
|------------|-------------------------|-----------------------|--------------------------|------------------------|
| 5          | ~ <del>₦142,500</del>   | ~ <del>₦50,280</del>  | ~ <del>₦300k–₦380k</del> | ~ <del>₦60k–₦76k</del> |
| 20         | ~ <del>₦570,000</del>   | ~ <del>₦120,000</del> | ~ <del>₦780k–₦870k</del> | ~ <del>₦48k–₦58k</del> |
| 50         | ~ <del>₦1,425,000</del> | ~ <del>₦250,000</del> | ~ <del>₦1.9M–₦2.0M</del> | ~ <del>₦43k–₦55k</del> |
| 100        | ~ <del>₦2,850,000</del> | ~ <del>₦420,000</del> | ~ <del>₦3.6M–₦3.8M</del> | ~ <del>₦40k–₦52k</del> |

Chatwoot stays flat per business (\$19/seat). AWS cost per business drops as you scale — the database, Redis, and Lambda are shared infrastructure. The more businesses on the platform, the cheaper it gets per business.

## 7. MVP vs. Production — 100 Users Per Business

| Metric                      | MVP (5 msgs/day)            | Production (100 msgs/day)               |
|-----------------------------|-----------------------------|-----------------------------------------|
| Messages/month per business | ~150                        | ~3,000                                  |
| AI token usage              | ~1.4M tokens                | ~28M tokens                             |
| Bedrock AI cost             | ~ <del>₦6,885</del>         | ~ <del>₦137,000</del>                   |
| Chatwoot per business       | ~ <del>₦28,500</del> (flat) | ~ <del>₦28,500</del> (same — flat rate) |
| Cost per business / month   | ~ <del>₦60k–₦76k</del>      | ~ <del>₦190k–₦200k</del>                |

The good news: Chatwoot is flat rate and doesn't increase with message volume. The only cost that scales with usage is AI (Bedrock). Once live, we can reduce AI costs by 30–50% by caching common answers and using lighter models for simple intents.

## 8. Next Steps

- Approve the ~~₦3.5M~~ build scope and agree on payment
- Kick off Milestone 1 — Foundation: backend, database, WhatsApp connection via Meta Embedded Signup
- Onboard first 5 pilot businesses for testing during Milestone 3 and 4

Each new business added after launch costs approximately ~~N60,000~~ ~~N76,000~~/month to run. Your subscription price to each business should sit comfortably above that. The platform is designed to be profitable from business #1.